A Comparative Study of Semiconductor Virtual Metrology Methods and Novel Algorithmic Framework for Dynamic Sampling

Xu Han, Marcella Miller, James Moyne, *Member, IEEE*, Gregory William Vogl, Anita Penkova, and Xiaodong Jia

***Abstract*—Virtual metrology (VM) is an important technology in semiconductor manufacturing that enhances process control, reduces costs, and improves quality. However, as processes become more complicated and process variations increase due to high-mix manufacturing, VM still faces challenges such as sensor drift and shift, as well as limited availability of metrology data due to high costs. This paper proposes an online Gaussian process (OGP) model designed to operate effectively with minimal initial metrology data and adapt dynamically to new data. The OGP model incorporates uncertainty quantification to optimize the sampling process, thereby enabling an adaptive sampling strategy to conduct metrology based on the process control needs. The proposed method is validated using a public dataset from the chemical mechanical planarization (CMP) process, demonstrating its effectiveness in tracking data drift and shift while reducing the required metrology data to retain model performance in an online operation setting.**

***Index Terms*—Semiconductor manufacturing, virtual metrology, uncertainty quantification.**

I. Introduction

As THE miniaturization of semiconductors continues, metrology and process (and quality) control of semiconductor manufacturing becomes increasingly difficult. Virtual metrology (VM), as a cost-effective alternative or augmentation to in-line metrology, has gained increasing importance in advanced process control (APC), yield management, chamber matching, etc. [1], [2].

Building a robust VM model requires extensive information from the processing chamber, but sensory data may experience drift and shift due to machine degradation and maintenance. In high-mix manufacturing, variations from different tools and recipe settings further complicate VM modeling. Consequently, VM models need to be frequently rebuilt with new recipes or after maintenance. Another major challenge is the limited availability of metrology data. While high-run recipes can have extensive historical data to train satisfactory VM models, low-run recipes (or new recipes) often struggle to develop accurate VM models due to the lack of metrology data or the lack of time and resources to support VM modeling from the design of experiments (DoE) [3].

Various machine learning and statistical techniques have been developed for VM applications. In practice, the exponentially weighted moving average (EWMA) is commonly used for run-to-run (R2R) control and APC [4], while a just-in-time adaptive disturbance estimation is proposed to tackle drift and shift issues directly [5]. For metrology prediction, techniques such as multivariate linear regression (MLR) [6], partial least squares [3], support vector regression [7], and neural networks [8] have been widely studied. Despite the diversity in these algorithms, they typically require substantial metrology data to build initial models. Moreover, when data shifts or drifts occur, these models often need to be retrained to maintain their accuracy and effectiveness. There is a lack of methods that effectively handle scenarios with limited initial metrology data.

In this paper, an online Gaussian process (OGP) model is proposed as a solution. The OGP model can be trained with few samples but improved continually through online learning as each new measurement is available, tracking the data shifts and drifts introduced by machine condition changes, maintenance, high mix operations, etc. To further minimize the metrology burden, the uncertainty quantification (UQ) of the OGP is leveraged to develop a preliminary justification of a dynamic sampling (DS) strategy. A comparative study is conducted to benchmark the performance of the OGP against other VM models. The effectiveness of the tracking ability of the OGP is demonstrated through three simulated scenarios using a public dataset from the chemical mechanical planarization (CMP) process.

The remainder of this paper is organized as follows: Section II reviews related work on VM and dynamic sampling. Section III details the proposed methodology and the benchmarked VM models. Section IV presents experimental results

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Xu Han is with the Department of Mechanical and Materials Engineering, University of Cincinnati, Cincinnati, OH 45221 USA (e-mail:hanx7@mail.uc.edu).

Marcella Miller is with the Department of Mechanical and Materials Engineering, University of Cincinnati, Cincinnati, OH 45221 USA.

James Moyne is with the Robotics Department, University of Michigan at Ann Arbor, Ann Arbor, MI 48109 USA.

Gregory William Vogl is with the Engineering Laboratory, National Institute of Standards and Technology, Gaithersburg, MD 20899 USA.

Anita Penkova is with the Aerospace and Mechanical Engineering Department, Viterbi School of Engineering, University of Southern California, Los Angeles, CA 90089 USA.

Xiaodong Jia is with the Department of Mechanical and Materials Engineering, University of Cincinnati, Cincinnati, OH 45221 USA.

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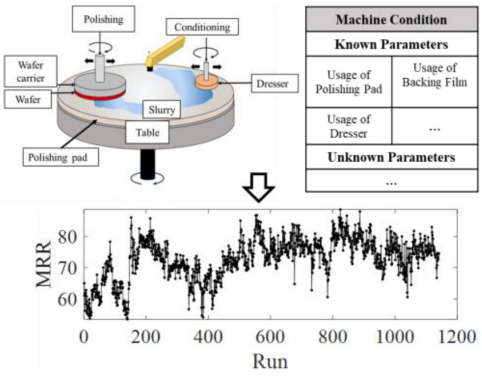


Fig. 1. The drift and shift in the MRR in the CMP process.

using the CMP dataset. Section V concludes and discusses future work.

II. Literature Review

Virtual metrology has been widely adopted in semiconductor manufacturing as an alternative to costly physical metrology in APC [1], [2], yield management [9], and chamber matching [10], [11], to reduce process costs and cycle times [12]. Li et al. [13] employed PCA for chamber wall deposit estimation in the dry etching process. Choi and Hong [14] developed a neural network-based VM model to predict the film thickness of an amorphous carbon layer in the deposition process. In the CMP process, where material removal rate (MRR) is critical, methods like the integrated VM model [15], and group method of data handling-based neural network [16], are developed for MRR prediction and identifying the key features and process parameters.

Despite the various VM models proposed for different processes, tracking data drift and shift remains a challenge, with causes including maintenance, sensor aging, and process variability [17]. To deal with the dynamic characteristics of the process, Khan et al. [3] employed recursive partial least squares within a feedback control loop in R2R to improve model performance, while Feng et al. [18] and Cai et al. [19] used online sample selection and a just-in-time particle filter, respectively, to dynamically adjust for changes in CMP.

In fact, wafer metrology is usually costly as it increases process cycle time. To reduce the required metrology data, a variety of sampling strategies are proposed to select wafers for metrology. Sequential sampling [20] and induced start DS [21] have been proposed to minimize the number of measured sites at each process run. Kang and Kang investigated a neural network-based dynamic sampling strategy for VM models, developing a reliability score using an ensemble neural network [8]. Susto developed a confidence level-based dynamic sampling method using random forest for the etching process [22]. The disadvantage of these methods is the substantial amount of data required for model training. However, both methods employ the concept of confidence level or UQ in VM models. As proven in [23], the predictive uncertainty of probabilistic models is related to the measurement uncertainty of physical inspection, making probabilistic models an excellent choice for VM applications. In some more recent work, Cai et al. proposed a reference-based strategy that incorporates uncertainty evaluation to enhance prediction reliability [24]. Zabrocki et al. proposed a novel adaptive online time-series learning to tackle the data drifts and shifts [25]. However, few papers considered using UQ in a DS scheme, especially under data drift and shift.

Based on the literature review, the following unmet needs in virtual metrology practices have been identified: 1) the need to compensate for partially known machine conditions, 2) the need to track data shifts and drifts, and 3) the need for a more flexible dynamic sampling strategy to reduce the metrology load. To address these challenges, this paper conducts a comprehensive benchmarking of seven different learning techniques. We propose an online Gaussian process (OGP) method that effectively tracks data shifts and drifts when metrology data is limited and provides uncertainty quantification (UQ) to support dynamic sampling strategies. The proposed methodology is validated using a real dataset from semiconductor fabrication in the Chemical Mechanical Planarization (CMP) process.

TABLE I
The Top Selected Features for MRR Prediction

Data Group	Top Sensors	Top Features	|CC|
1	Chamber pressure	Area Under Trace	0.62
	Usage of polishing table	Standard Deviation	0.53
	Usage of dresser table	Mean	0.51
	Usage of membrane	Standard Deviation	0.51
	Usage of backing film	Area Under Trace	0.49
	Usage of Dresser	Mean	0.44
2	Slurry flow C	Area Under Trace	0.74
	Main outer air bag pressure	Area Under Trace	0.72
	Chamber pressure	Area Under Trace	0.67
	Usage of dresser table	Mean	0.61
	Usage of polishing table	Standard Deviation	0.59
	Usage of Dresser	Mean	0.34

III. Methodology

A. Virtual Metrology

VM provides a cost-effective and time-efficient alternative to physical metrology. The real-time model prediction allows R2R controllers to adjust process settings for the next wafer run based on the predictive information from the current run. However, the wide adaptation of VM in production has several challenges, leading to three critical needs.

1) Need 1 (Modeling the Partially Known Machine Condition): Fig. 1 shows a schematic diagram of the CMP process. The MRR exhibits significant drift and shift, posing challenges for VM modeling. In addition to the control process inputs, the machine's condition at different times also influences the obtained MRR. Certain known conditions can be monitored by sensors, such as the usage of consumable components like the polishing pad, dresser, and dresser table. These variables have been extensively studied in prior works. Some features with high correlation with MRR, as listed in Table I, can be extracted from

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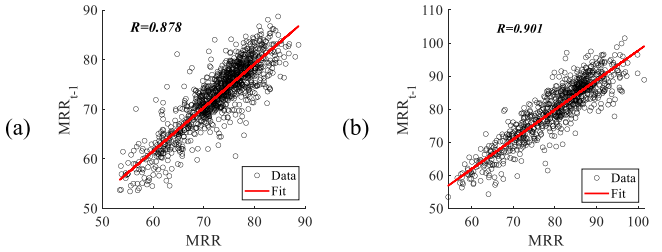


Fig. 2. Correlation of MRR at time t with MRR at time $t-1$ for (a) Data Group 1, and (b) Data Group 2 for CMP dataset.

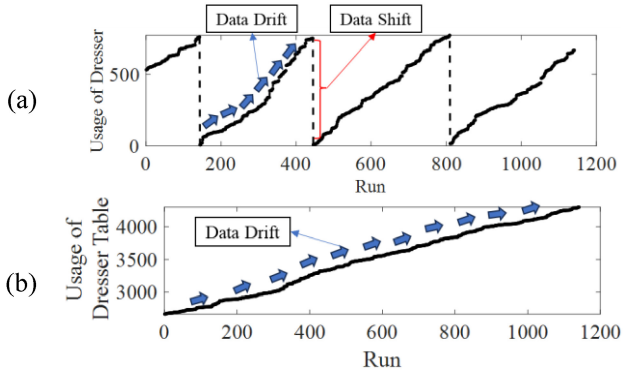


Fig. 3. Drift and shift in (a) usage of dresser and (b) usage of dresser table.

process inputs and usage variables to build a basic VM model.

However, VM models developed using only these features are often insufficient, as usage variables only represent part of the machine conditions. There are also unknown parameters that cannot be monitored. Despite this, it is reasonable to hypothesize that the machine condition remains relatively unchanged over short periods. This hypothesis is supported by the high correlation between temporally adjacent MRR values. As shown in Fig. 2, the high correlation between adjacent MRR measurements in both data group 1 and 2 can compensate for the partially known machine condition in the model. Therefore, including a dynamic term, specifically the time-adjacent reference metrology, can improve the model's tracking ability.

2) Need 2 (Tracking Data Drift and Shift): Data drift and shift occur not only in the MRR but also in the usage variables. Consumable components like the polishing pad, dresser, and dresser table degrade over time, reducing polishing effectiveness and introducing disturbances when replaced during maintenance, which leads to inaccurate predictions. Fig. 3 shows the drift and shift in the usage of the dresser and the dresser table. A common approach to address these changes is error compensation, where large prediction errors are tracked with a moving average and added to the current prediction to adapt to metrology dynamics. Alternatively, model parameters can be adapted to feature changes. An initial VM model can be built with a small amount of data, and after deployment, the model parameters can be updated online to improve performance as more samples are included.

3) Need 3 (Fixed-Rate Sampling and Dynamic Sampling): To update the model parameters online, metrology data must be collected. However, most measurements are costly in time, making it impractical to measure every wafer. Therefore, a sampling scheme to reduce the metrology load is needed. The common strategies are fixed rate or random sampling, where predetermined or random wafers from a lot are selected for measurement. Although consistent in providing sufficient metrology data to monitor the process, these strategies do not consider the model's prediction confidence in decision-making and instead use random or preselected samples regardless of the VM.

A more intelligent strategy is dynamic sampling, which considers prediction uncertainty. When the model is confident in its current prediction, it is reasonable to skip sampling the current wafer, further reducing the metrology load. The basis of DS is the model's ability to quantify uncertainty. This paper compares a series of conventional VM models with different capabilities in tracking, online model updating, and dynamic sampling. We propose an OGP model to address the aforementioned VM needs, particularly when very limited wafer metrology data is available.

TABLE II
BENCHMARKING ALGORITHMS

VM Method		
MLR	$\hat{y}_k = \boldsymbol{\theta}^T \cdot \boldsymbol{u_k} + e_k$	(1)
VM Methods with Tracking Ability		
EWMA	$\hat{y}_k = \boldsymbol{\theta}^T \cdot \boldsymbol{u_k} + e_k$ $e_k = \alpha(y_{k-1} - \boldsymbol{\theta}^T \cdot \boldsymbol{u_{k-1}}) + (1-\alpha)e_{k-1}$	(2)
ARX	$\hat{y}_k = a \cdot y_{k-1} + \boldsymbol{\theta}^T \cdot \boldsymbol{u_k} + e_k$	(3)
KF	$\begin{cases} x_k = a \cdot x_{k-1} + \boldsymbol{\theta}^T \cdot \boldsymbol{u_k} + p_k \\ y_k = x_k + q_k \end{cases}$	(4)
VM Methods with Online Model Updating		
DLKF	*KF for state estimation:* $\begin{cases} x_k = a_k \cdot x_{k-1} + \boldsymbol{\theta_k^T} \cdot \boldsymbol{u_k} + p_k \\ y_k = x_k + q_k \end{cases}$	(5)
	KF for parameter estimation: $\begin{cases} \boldsymbol{w}_k = \boldsymbol{w}_{k-1} \\ y_k = \boldsymbol{w_k^T} \cdot \begin{bmatrix} x_{k-1} \\ \boldsymbol{u_k} \end{bmatrix} + p_k + q_k \end{cases}$	(6)
BARX	$\hat{y}_k = a_k \cdot y_M + \boldsymbol{\theta_k^T} \cdot \boldsymbol{u_k} + e_k$	(7)
OGP	*GP Model:* $\hat{y}_k = GP(m_k, \Sigma_k) + e_k, \quad m_k = a_k \cdot y_{k-1}$	(8)
	Prediction Function: $\hat{y}_k = a_k \cdot y_M + \Sigma_k(u_*, U)\Sigma_k^{-1}(y - m)$	(9)

B. Analytics for Virtual Metrology

In this work, the OGP model is benchmarked against six other commonly used algorithms for virtual metrology, including MLR, EWMA, auto-regression with exogenous variable (ARX), Kalman filter (KF), Bayesian auto-regression with exogenous variable (BARX), and dual linear Kalman filter (DLKF). The algorithms are selected with an increasing model complexity and enhanced capability. The equations for prediction models are tabulated in Table II.

The traditional VM methods, like MLR, model the relationship between predictors and metrology using a static model, as in Eq. (1), where $\hat{y}_k$ is the prediction of the VM, $\boldsymbol{\theta}^T$ represents

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the state model parameters, which can be obtained from the DoE, $\boldsymbol{u_k}$ is the feature set extracted from the process, and e_k is the error term. Once the training process is completed, the parameters are fixed, making the model vulnerable to dynamic processes where sensors are subject to drift and shift.

To address this problem, a dynamic term can be introduced, transforming the VM model into a dynamic model. This term, often a recent measurement from the same machine, helps track drift and shift in metrology and compensates for partially known machine conditions. This can be achieved in through various methods, such as error correction used in EWMA in Eq. (2), direct modeling of an exogenous variable in ARX in Eq. (3), and the state modeling employed in KF in Eq. (4).

Note that in the EWMA, ARX, and KF models, the state parameter $\boldsymbol{\theta^T}$ remains static. In scenarios where the MRR is affected by long-term system drift due to degradation or material usage changes, the dynamic term alone may not suffice for maintaining prediction accuracy. The model may be subject to retraining after a maintenance event. To address this in the VM model, a time-variant Kalman filter, namely DLKF, is proposed by the authors in a recent work [26], as shown in Eq. (5) and (6). In DLKF, a_k and $\boldsymbol{\theta}_k^{\boldsymbol{T}}$ remain as weights of the recent metrology and state model, but the subscript k suggests their dynamic nature. Note that the state vector $\boldsymbol{w}_k = \begin{bmatrix} a_k & \boldsymbol{\theta}_k \end{bmatrix}^T$ in Eq. (6) is a collection of the model parameters and the noise terms $p_k \sim N(0, \sigma_p^2)$ and $q_k \sim N(0, \sigma_q^2)$.

Online Updating for DLKF: The implementation of DLKF has three steps – state prediction, state update, and parameter update – which are elaborated in equations (10)–(12). The state and model parameters are updated when a metrology point is available. $H_k = \begin{bmatrix} x_{k-1} & \boldsymbol{u_k} \end{bmatrix}^T$ is the first derivative of the state equation in Eq. (6) at time k.

State Prediction:

$$\begin{aligned} x_k^- &= a_{k-1} \cdot x_{k-1} + \boldsymbol{\theta}_{k-1} \cdot \boldsymbol{u_k} \\ P_{x,k}^- &= a_{k-1} \cdot P_{x,k-1} \cdot a_{k-1}^T + \sigma_p^2 \end{aligned} \tag{10}$$

State Update:

$$\begin{aligned} K_k &= P_{x,k}^- \cdot \left(P_{x,k}^- + \sigma_p^2 \right)^{-1} \\ P_{\widehat{\boldsymbol{x}},k} &= (I - K_k) \cdot P_{x,k}^- \\ x_k &= x_k^- + K_k \cdot (y_k - x_k^-) \end{aligned} \tag{11}$$

Parameter Update:

$$\begin{aligned} \boldsymbol{w}_k^- &= \boldsymbol{w}_{k-1} \\ P_{w,k}^- &= P_{w,k-1} \\ K_{w,k} &= P_{w,k}^- \cdot H_k^T \cdot \left(H_k \cdot P_{w,k}^- \cdot H_k^T + \sigma_p^2 + \sigma_q^2 \right)^{-1} \\ P_{\widehat{\boldsymbol{w}},k} &= \left(I - K_{w,k} \cdot H_k \right) \cdot P_{w,k}^- \\ \boldsymbol{w}_k &= \boldsymbol{w}_k^- + K_{w,k} \cdot \left(y_k - \boldsymbol{w}_k^{-\boldsymbol{T}} \cdot \begin{bmatrix} x_{k-1} \\ \boldsymbol{u_k} \end{bmatrix} \right) \end{aligned} \tag{12}$$

While it manages to adapt to data shifts and drifts, DLKF has two limitations. DLKF requires sufficient data to build a robust initial state model and only supports fixed-rate sampling, which makes it less effective in scenarios where initial training samples are limited or dynamic sampling is needed.

Online Updating for BARX: As an alternative, BARX can sequentially update the ARX model parameters once new metrology data is obtained. Unlike the ARX model, BARX updates the model parameters recursively using Bayesian rules whenever a new observation is available. Additionally, the predictive variance serves as a metric for uncertainty quantification. This capability allows BARX to be adapted to a DS scheme, where metrology data is only collected when the model's prediction is uncertain, thereby reducing the number of metrology samples and increasing throughput. Consequently, the reference measurement y_M used in Eq. (7) does not need to be y_{k-1} as in ARX; it can be any recent available measurement.

However, a limitation of the online updating rules for BARX [18] is the lack of a forgetting mechanism. As more samples are included during online updating, the predictive variance converges, diminishing its ability to quantify uncertainty. To address this issue, a kernel-based model with superior uncertainty quantification can be utilized.

Online Updating Method for GP: The prediction function of the OGP is shown in Equation (13). When a new measurement ($\boldsymbol{u_+}$, y_+) is available, the model can be updated in one of two ways. The first approach, as described in Equations (14)-(15), involves only online kernel augmentation. In this step, the covariance matrix $\boldsymbol{\Sigma}_{**}^+$ and mean function $\boldsymbol{m}_*^+$ of the GP model are updated based on the new data. In this work, the squared exponential (SE) kernel is used as described in Eq. (16), where σ_f^2 is the variance and l is the length scale of the kernel. During this step, the kernel matrix is augmented to include the new data, while the kernel parameters remain unchanged. After augmentation, the uncertainty quantification is updated to reflect the influence of the most recent metrology data. This approach maintains the efficiency of the GP by incorporating new information without recalculating the entire model, which would be computationally expensive.

OGP Prediction:

$$f_* \sim GP\left(\boldsymbol{m}_*^+, \ \boldsymbol{\Sigma}_{**}^+\right) \tag{13}$$

Kernel Augmentation:

$$\begin{aligned} \boldsymbol{\Sigma}_{**}^+ = \boldsymbol{K}_{**} - \begin{bmatrix} K_{*N} & K_{*+} \end{bmatrix} \Bigg(\begin{bmatrix} K_{NN} & K_{N+} \\ K_{+N} & K_{++} \end{bmatrix} \\ + \begin{bmatrix} \widehat{\Sigma}_{f_N} & 0 \\ 0 & \sigma_{f_+}^2 \end{bmatrix} \Bigg)^{-1} \begin{bmatrix} K_{N*} \\ K_{+*} \end{bmatrix} \end{aligned} \tag{14}$$

$$\begin{aligned} \boldsymbol{m}_*^+ = \boldsymbol{m}_* + \begin{bmatrix} K_{*N} & K_{*+} \end{bmatrix} \Bigg(\begin{bmatrix} K_{NN} & K_{N+} \\ K_{+N} & K_{++} \end{bmatrix} \\ + \begin{bmatrix} \widehat{\Sigma}_{f_N} & 0 \\ 0 & \sigma_{f_+}^2 \end{bmatrix} \Bigg)^{-1} \left(\begin{bmatrix} \boldsymbol{f}_N \\ f_+ \end{bmatrix} - \begin{bmatrix} \boldsymbol{m}_N \\ m_+ \end{bmatrix} \right) \end{aligned} \tag{15}$$

$$\text{where } K_{ij} = K\left(\boldsymbol{x}_i, \ \boldsymbol{x}_j\right) = \sigma_f^2 \exp\left(-\frac{\left(\boldsymbol{x_i} - \boldsymbol{x_j}\right)^2}{2\boldsymbol{l}^2} \right) \tag{16}$$

Kernel Parameters Update:

$$\begin{aligned} NLML &= -logp(\boldsymbol{y}|\boldsymbol{X}) \\ &= \frac{1}{2} y^T (\Sigma + \sigma_n^2 I)^{-1} y + \frac{1}{2} log|\Sigma + \sigma_n^2 I| + \frac{n}{2} log 2\pi \end{aligned} \tag{17}$$

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TABLE III
SUMMARY OF KEY NOTATIONS

y_{k-1}, u_k, x_k	Reference metrology, control variables, state variable
$a_k, \boldsymbol{\theta}^T, \boldsymbol{w}_k^T$	Weights for reference metrology, control variables, and DLKF parameters
p_k, q_k	Process noise and observation noise
m_k, Σ_k	Posterior mean and covariance of GP function
$\boldsymbol{m}_*^+, \boldsymbol{\Sigma}_{**}^+$	Posterior mean and covariance of GP function after including new measurement
$\boldsymbol{m}_*, \boldsymbol{K}_{**}$	Mean and covariance between test points
$\boldsymbol{K}_{*N}, K_{*+}$	Covariance between test points and training points, and test points and new measurement
$\sigma_{f_+}^2, \boldsymbol{l}$	Variance and length scale of kernel function

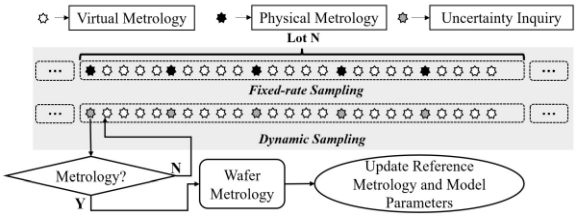


Fig. 4. Proposed dynamic sampling scheme.

$$\boldsymbol{\Theta} = \arg\min_{\boldsymbol{\Theta}} -logp(\boldsymbol{y}|\boldsymbol{X}), \text{ where } \boldsymbol{\Theta} = \left[\sigma_f^2\ \boldsymbol{l}\right]^T \quad (18)$$

The second scheme involves minimizing the negative log marginal likelihood (NLML), which updates the kernel parameters based on the current data available. The NLML can be expressed by Eq. (17). Since the NLML is a convex function, it can be optimized using the conjugate gradient method, as shown in Eq. (18). The computational complexity of kernel augmentation is $O(n)$, and the complexity of kernel parameter updates is $O(n^2)$. In practical implementation, updating the kernel parameters is not required after every metrology measurement. Depending on the available computational resources, the quality control requirements, and the metrology scheduling for different recipes, kernel parameter updates can be performed after each batch or after several batches. Table III summarizes some of the key notations used across different VM algorithms.

C. Methodology for Dynamic Sampling

Metrology in semiconductors is normally time-consuming, and some sampling strategy is needed to select wafers for metrology. Fixed-rate sampling is commonly employed in practice as a systematic approach to metrology due to its simplicity and consistency. This strategy involves selecting wafers at regular, predetermined intervals, such as the first of every five wafers from each lot, as shown in Fig. 4. Fixed-rate sampling ensures consistent and even coverage across the lot, reducing the metrology demand effectively while maintaining process control and quality assurance. However, with this method, some of the measurements could be redundant if the model is well-trained in the current feature space.

To reduce the metrology data, we propose a simple DS method enabled by the uncertainty quantification capabilities of the BARX and OGP algorithms. As an extension of fixed- or random-sampling schemes, this method incorporates a decision step based on the model's predictive variance to determine whether a sample requires metrology. Low predictive variance indicates high confidence in the model's predictions, suggesting that the feature space is well-explored and learned. Conversely, high predictive uncertainty indicates the need for additional metrology to update the model. A dynamic threshold, T, defined as the tth percentile of predictive variance across samples, is applied in the decision step to assess metrology needs. The initial value of T is established through a leave-one-out test on the training set.

In practical implementation, varying the variance threshold percentile t allows for the generation of an empirical graph that illustrates the trade-off between sampling cost and model performance. The optimal t can then be chosen through a more complex and comprehensive decision-making process that accounts for factors such as equipment availability, personnel workload, and quality control requirements.

While many DS schemes exist, this proposed method demonstrates the effectiveness of the algorithms' uncertainty quantification. The OGP algorithm and DS scheme can be integrated into a more comprehensive decision-making system as needed.

TABLE IV
DESCRIPTION OF TRAINING AND TESTING DATASET

Data Group	Training	Testing	Total
1	100	1041	1141
2	100	1067	1167

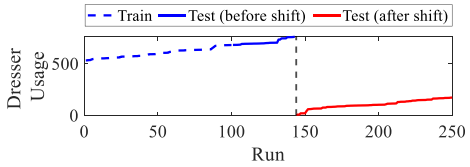


Fig. 5. Settings for algorithm tracking ability evaluation.

IV. RESULTS AND DISCUSSION

A. Case Study I: Data Tracking Ability Assessment

The first case study aims to evaluate the tracking ability of the selected algorithms shown in Table II. The entire dataset is sorted by the cycle start time and then split into training and testing sets, as detailed in Table IV. To assess the performance of the selected algorithm in a relatively realistic setting, the training data comprises only the first 100 runs, as the initial VM model is typically established based on a limited number of observations from the start of the experiment. A visualization of the first 250 runs in data group 1 is shown in Fig. 5, where the first 100 runs are set as training data. It is worth noting that two variables are experiencing drift and shift, and we selected one occurrence to test the models' performance with unseen data. Table V summarizes the Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) for predictions from all the benchmarked models.

The large prediction errors indicate that MLR struggles to adapt to data shifts and drifts, particularly noticeable at the

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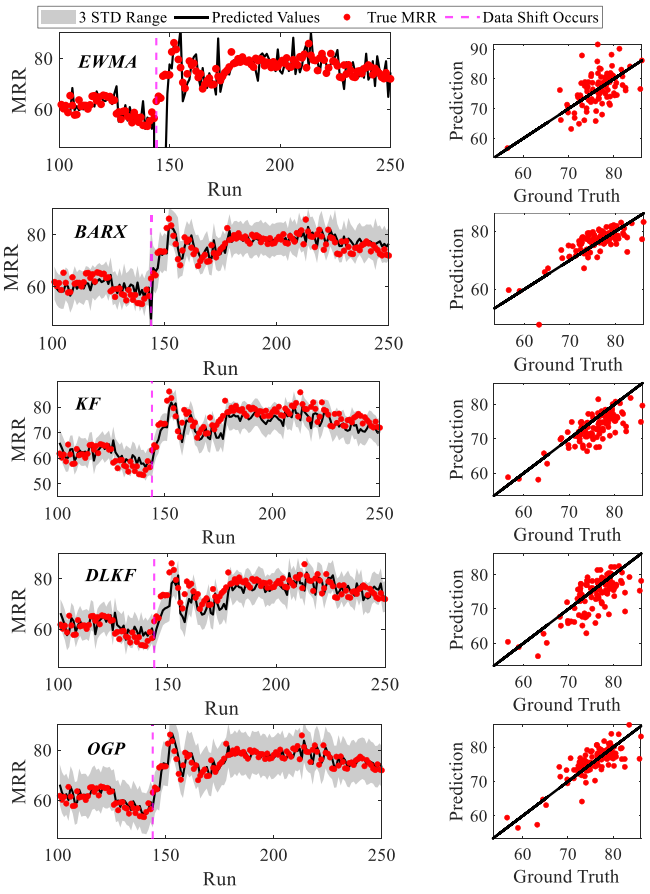


Fig. 6. Visualizing the tracking ability of selected algorithms.

144th wafer run when a disturbance occurs. Similar prediction errors are seen with the ARX and EWMA models due to their reliance on the most recent metrology value y_{t-1}, as shown in Eqs. (2) and (3). EWMA is often favored in practice because its weighting factor for recent measurements can be manually adjusted. This flexibility allows users to fine-tune the tracking ability of the algorithm by designing either a fast or slow EWMA. In contrast, the ARX model decides this weighting through model training. This inherent difference eliminates the trial-and-error effort involved in tuning the α coefficient in EWMA, but it results in a loss of flexibility. The KF further modifies the ARX model using a prediction-correction scheme to improve tunability. The KF predictions show slightly worse results than the ARX model. However, by adjusting the variances of the state noise and observation noise, KF can achieve results very similar to those of the ARX model.

Fig. 6 provides a detailed visualization of all selected algorithms. The results show that EWMA struggles to track the MRR after the 144th wafer run where a disturbance is introduced due to the replacement of the dresser. However, this issue can be mitigated by adjusting the EWMA parameter to improve its ability to track fast changes. Another limitation of EWMA is its inability to provide uncertainty quantification, making it unsuitable for DS where UQ is necessary to determine when wafer runs require metrology. In contrast, BARX, KF, and OGP demonstrate satisfactory tracking abilities and provide the needed UQ.

TABLE V
BENCHMARKING RESULTS OF ALL MODELS

Data Group/ Models	MSE			MAPE		
	1	2	Total	1	2	Total
MLR	1148.78	144.48	640.43	38.83	12.72	25.62
ARX	15.12	32.49	23.91	4.13	5.13	4.64
EWMA	14.71	24.43	19.63	3.65	3.97	3.81
KF	18.52	33.19	25.95	4.74	5.21	4.98
DLKF	14.79	34.8	25	3.96	4.57	4.27
BARX	12.59	20.08	16.41	3.85	4.31	4.09
OGP	**8.73**	**14.29**	**11.57**	**3.01**	**3.2**	**3.11**

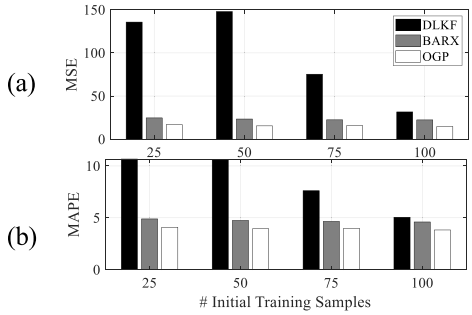


Fig. 7. Influence of initial training data size on prediction outcome in terms of (a) MSE and (b) MAPE.

B. Case Study II: Fixed-Rate Sampling

The second case study evaluates the algorithms in a fixed-rate sampling setting. Normally, the first and last wafers in a lot are sampled for metrology. During the ramp-up of a new product (or recipe), the sampling rate might increase. In this case study, we first benchmark the influence of the size of the initial training data. The results in Fig. 7 indicate that BARX and OGP demonstrate robustness to the size of the initial training data. Subsequently, we chose 25 wafers as the initial training dataset to evaluate the tracking abilities of BARX, DLKF, and OGP.

Fig. 8 presents the benchmarking results for BARX, DLKF, and OGP, indicating that DLKF has a poor tracking ability when the initial training set is small, requiring a long tune-in period for effective tracking. Additionally, DLKF requires a fixed sampling rate and cannot be utilized for dynamic sampling. In contrast, BARX and OGP yield satisfactory results in terms of both UQ and tracking ability. Table VI shows results with varied sampling rates for an initial training set of 50 wafers.

C. Case Study III: Dynamic Sampling

To support a DS strategy, the model must be able to update parameters at arbitrary steps, which disqualifies DLKF, as it can only be updated at fixed intervals. Therefore, in the third case study, we compare the proposed OGP model with BARX to demonstrate its effectiveness under the DS scheme. Metrology inquiries are incorporated into samples selected from Case Study II, with metrology required at points where predictive uncertainty exceeds the threshold T. Fig. 9 shows

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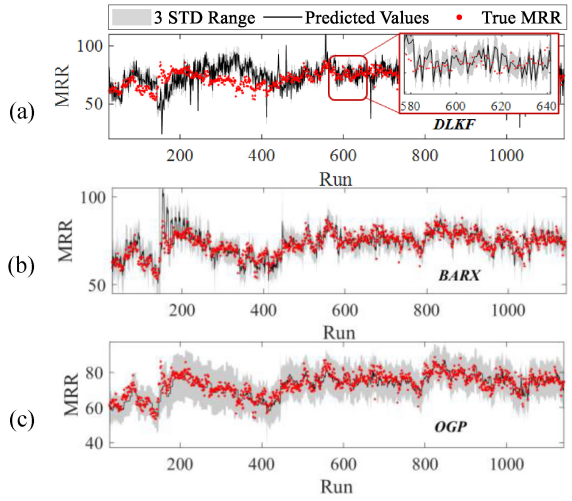


Fig. 8. Predictions for (a) DLKF, (b) BARX, and (c) OGP. The initial training set has 25 wafer runs, and five samples are drawn from each lot.

TABLE VI
THE DETAILED BENCHMARKING OF DLKF, BARX, AND OGP

Data group	# Samples per lot	MSE			MAPE		
		DLKF	BARX	OGP	DLKF	BARX	OGP
1	25	30.90	10.56	**8.92**	5.74	3.35	**3.05**
	12	40.00	13.14	**11.55**	6.73	3.80	**3.48**
	5	62.03	21.90	**14.17**	8.34	4.75	**4.00**
	3	89.64	25.65	**15.07**	10.09	5.38	**4.16**
	2	115.37	60.95	**23.44**	11.65	8.37	**5.01**
	1	191.71	66.24	**28.77**	15.41	7.86	**5.39**
2	25	208.85	15.40	**13.50**	11.37	3.67	**3.17**
	12	210.27	20.60	**13.68**	11.48	4.08	**3.44**
	5	231.87	24.83	**16.68**	12.75	4.75	**3.90**
	3	245.92	48.30	**20.45**	13.29	6.06	**4.36**
	2	251.14	56.28	**27.66**	14.07	7.41	**5.22**
	1	318.49	84.80	**40.39**	16.00	8.55	**6.12**

the predictive variance of BARX and OGP when 50 initial samples are used to train the model. When the disturbance is introduced at the 144th wafer run, both models exhibit large prediction uncertainties, indicating a lack of confidence in their predictions. Consequently, more metrology data is collected initially. Due to the online learning capability of both models, once a sufficient number of measurements is collected to train a robust model, the predictive uncertainties mostly fall below the threshold, thereby reducing the metrology costs in time.

Fig. 10 presents the MSE and MAPE when different variance threshold percentiles are used. When the threshold is set to 0, the scheme is effectively a fixed-rate sampling as shown in case II. As the threshold increases, the total number of required measurements starts to decrease, while maintaining low MSE and MAPE. Eventually, when the threshold is set to 1, the models degenerate due to the lack of samples for online learning. This comparison reveals that the proposed OGP is significantly better than BARX for every threshold. When the threshold is set to 0.8, the OGP reaches a balance between the number of samples required and the model accuracy.

With the threshold set to 0.8, three kernels—Square Exponential (SE), Rational Quadratic (RQ), and Matern—are selected and benchmarked against BARX. The MSE, MAPE,

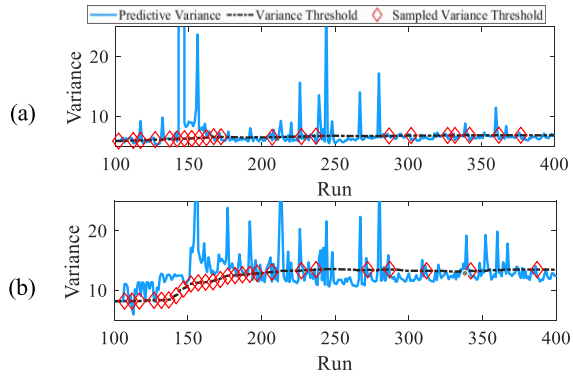


Fig. 9. Model predictive variance and sampled data for (a) BARX and (b) OGP at $t = 0.8$ (Variance Threshold) with 50 initial training samples.

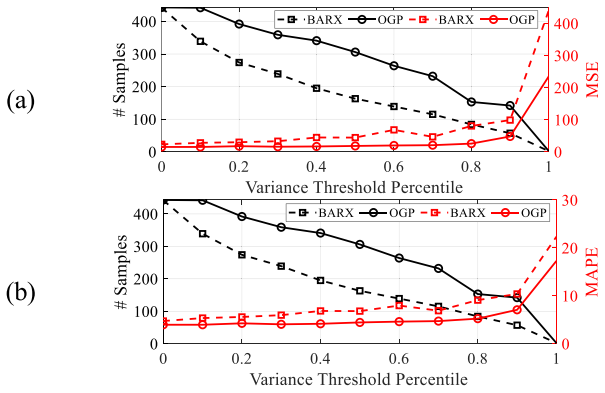


Fig. 10. Number of samples vs. model performance under different variance thresholds for (a) BARX and (b) OGP.

TABLE VII
BENCHMARK OF DIFFERENT KERNEL SELECTION

Data Group/Models	MSE		MAPE		# Samples	
	1	2	1	2	1	2
BARX	49.32	111.11	7.56	10.59	**53**	**31**
OGP (SE)	**19.40**	30.27	**4.76**	5.59	81	69
OGP (RQ)	20.44	**27.44**	4.98	**5.05**	92	102
OGP (Matern)	19.56	28.14	4.78	5.16	82	93

and the number of samples required under the DS scheme are tabulated in Table VII. Overall, the OGP demonstrates better model performance and UQ capability compared to the BARX model under the dynamic sampling scheme. The BARX model is hindered by the issue of vanishing variance, where the predictive variance diminishes rapidly as the number of training samples increases. This results in overly confident predictions and a loss of capability for effective UQ. In contrast, the kernel-based method employed by OGP avoids the vanishing variance problem, providing more reliable UQ and achieving a better balance between model performance and sampling cost. Notably, under fixed sampling, the total number of samples required is 88 and 92 for data groups 1 and 2, respectively. With the DS strategy, the OGP achieves superior performance with fewer or equal samples.

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V. Conclusion

This paper presents a comparative study of VM models and introduces an online Gaussian process (OGP) model with DS strategies for semiconductor manufacturing. Seven different algorithms for VM are benchmarked, with the OGP model demonstrating promising results, particularly when initial metrology data is limited. This allows better adaptation and transition from a DoE model to a VM model in R2R control. By incorporating dynamic sampling through uncertainty quantification, the OGP model effectively balances sampling cost and model performance, further reducing the metrology load while maintaining accuracy. This capability offers potential cost savings for semiconductor fabs. While the method demonstrates promising results, a remaining practical challenge, however, is the expansion of the kernel matrix as metrology data accumulates. Future work will explore a more efficient implementation of sparse online Gaussian processes to address the computation complexity. Other future investigations include updating the mean function and kernel parameters separately, potentially allowing for improved tracking under large disturbances. The dataset and code supporting this paper are available at: https://github.com/IntelligentMetrologySystems/Semiconductor_Virtual_Metrology.

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